

Determining Sample Size

Wenchao Ma



Why determining sample size?

- Estimating population parameters within a certain margin of error
- Conducting hypothesis testing
 - Whether two variables are correlated or not
 - Whether X can predict Y
- Impact of sample size
 - Small sample size →



Review: Hypothesis Testing

- **A hypothesis** is a statement or proposed explanation for an observation, a phenomenon, or a scientific problem that can be tested using the research method.
- A hypothesis is often a statement about the value for a parameter in a population.

An example

- You want to conduct a survey to check not only the prevalence of anxiety and depression in UMN undergraduates but also want to assess whether they are correlated.



Hypothesis Testing

- *Hypothesis Testing* - method for testing a hypothesis about a parameter of a population, using data measured in a sample
- To determine the likelihood of observing your data when the hypothesis is true



Step 1: State the null and alternative hypothesis

- Null Hypothesis (H_0 or H_N) - statement about the population parameter (such as the mean) that is **assumed to be true**
 - Example: anxiety and depression are NOT correlated
- Alternative hypothesis (H_1 or H_A) - statement that contradicts the null hypothesis
 - We think null is wrong, H_1 allows us to state what we think is wrong
 - Anxiety and depression are correlated



- How do we determine if the null hypothesis should be rejected?
 - Take a sample of UMN students.
 - Find the correlation r and compare it with 0.
 - If it is “too different” from 0, we reject the null hypothesis.
 - If it is “close enough” to 0, we retain the null hypothesis.

Statistical Power

- Probability, given that H_0 is false, that you will obtain outcomes in your sample large enough to reject H_0 .
- A probability, denoted $(1 - \beta)$,
where β = risk of Type II error.



Decision Error in Statistics

- Two major types of decision errors:
 - **Type I error:** We falsely reject H_0
 - *False positive*
 - **Type II error:** We decide not to reject H_0 when we should
 - *False negative*



Factors that Affect Statistical Decisions

- Sample size:
 - **Larger samples** make it more likely that we can reject H_0 (leading to Type II errors)
 - We will talk more about this later 😊
- Alpha
 - The **smaller** the value of α , the more difficult it is to reject H_0



Back to statistical power

- “Power” to detect an effect when there is one
- We want power to be reasonably high
 - If you want a guideline: Usually, about .80
 - Corresponds to keeping the risk of Type II error low

